

ON COMPUTATION OF WL-CLOSURES OF SMALL DEZA GRAPHS

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Based on the results obtained during The First Workshop at
the Mathematical Center in Akademgorodok

Novosibirsk, November 2020

Weisfeiler-Lehman algorithm

Weisfeiler-Lehman algorithm [1] (2-dim WL)

Input: graph Γ with vertex set V .

Output: WL-closure of Γ .

1. For each pair $(u, v) \in V \times V$ set $c(u, v) = 0, 1$ or 2 if (u, v) is a loop, edge or co-edge respectively.
2. Compute multiset $m(u, v) = \{(c(u, w), c(w, v)) : w \in V\}$.
3. Transform multiset $m(u, v)$ into a vector $s(u, v)$: the smallest pair is set to the beginning, the next smallest pair is set next, etc.
4. Define new coloring c' as follows:

$$c'(u, v) < c'(a, b) \Leftrightarrow c(u, v) < c(a, b) \text{ or } s(u, v) < s(a, b).$$

5. Repeat step 2 if $|c| \neq |c'|$. Otherwise return c .

[1] B. Weisfeiler, A. Leman, Reduction of a graph to a canonical form and an algebra which appears in the process, *NTI, Ser. 2*, **9** (1968) 12–16

Let $S^\cup$ be the set of all binary relations of WL-closure of Γ .
Relation e from $S^\cup$ is called **parabolic** if e is an equivalence.

Strongly regular graphs

A k -regular graph on v vertices is called a **strongly regular graph** with parameters (v, k, λ, μ) , if any two adjacent vertices in this graph have λ common neighbours and any two distinct non-adjacent vertices have μ common neighbours.

Deza graphs

Deza graphs were introduced in 1999 [2] as a generalisation of strongly regular graphs.

A k -regular graph on v vertices is called a **Deza graph** with parameters (v, k, b, a) , $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b .

A Deza graph is called a **strictly Deza graph** if it has diameter 2 and is not strongly regular.

[2] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Enumeration of strictly Deza graphs

In 1999 the complete list of strictly Deza graphs with at most 13 vertices was presented (see [2]).

In 2011 an algorithm for enumeration of Deza graphs was presented by S. Goryainov and L. Shalaginov and this list was extended up to 16 vertices (see [3]).

In 2019 this list was extended up to 21 vertices with new version of this algorithm by S. Goryainov, D. Panasenko and L. Shalaginov. In 2014 the complete list of Cayley-Deza graphs with $a > 0$ up to 59 vertices was presented by S. Goryainov and L. Shalaginov (see [4]).

[3] S. V. Goryainov, L. V. Shalaginov, On Deza graphs with 14, 15, and 16 vertices, *Sib. Electron. Mat. Izv.*, **8**, (2011) 105–115

[4] S. V. Goryainov, L. V. Shalaginov, Cayley-Deza graphs with less than 60 vertices, *Sib. Electron. Mat. Izv.*, **11**, (2014) 268–310

Noncirculant Cayley-Deza graphs with WL-rank 4

Parameters	Groups
(14, 9, 6, 4)	D_{14}
(18, 9, 8, 4)	$C_3 \times S_3; (C_3 \times C_3) : C_2; C_6 \times C_3$
(20, 7, 6, 2)	$C_5 : C_4$
(20, 13, 12, 8)	$C_5 : C_4$
(21, 8, 4, 2)	$C_7 : C_3$
(21, 12, 7, 5)	$C_7 : C_3$
(22, 16, 12, 10)	D_{22}
(24, 8, 4, 2)	$C_4 \times S_3; D_{24}; C_2 \times (C_3 : C_4); (C_6 \times C_2) : C_2;$ $C_{12} \times C_2; C_3 \times D_8; S_4; C_2 \times A_4; C_2 \times C_2 \times S_3;$ $C_6 \times C_2 \times C_2$
(24, 14, 8, 7)	S_4
(27, 18, 12, 9)	$(C_3 \times C_3) : C_3; C_9 : C_3$
(27, 18, 12, 9)	$(C_3 \times C_3) : C_3; C_9 : C_3$
...	...

Total: 27 graphs up to 59 vertices.

The table of Deza Graphs is available at:

<http://alg.imm.uran.ru/dezagraphs/dezatab.html>

The table of Cayley-Deza Graphs is available at:

<http://alg.imm.uran.ru/dezagraphs/dezatab.html>

The file with the implemented WL-algorithm is available at:

<http://alg.imm.uran.ru/wl.php>

Thanks for your attention!